

DATA CLEANING & PREPARATION

PDF Change Log — Project 1 Artifact
DecodeLabs Industrial Training Kit | Batch 2026

1,200

Total Records

14

Columns

6

Changes Applied

309

Nulls Imputed

0

Duplicates Found

0%

Date Format Errors

EXECUTIVE SUMMARY

This document is the official Change Log artifact for DecodeLabs Data Analytics Project 1. The raw dataset contained **1,200 order records** across 14 columns spanning 2023–2025. The cleaning pipeline was executed in **three phases** following the DecodeLabs methodology: **Phase 1 — Strategic Imputation** (handling missing values without data loss), **Phase 2 — Integrity Audit** (duplicate elimination and ID uniqueness verification), and **Phase 3 — Format Standardisation** (dates, numerics, and text normalisation). All 6 change records are documented below with full impact assessment. Final verification confirms **0% error rate on unique identifiers** and **0% error rate on date formats**, satisfying the Project 2 threshold gate.

✓ 0 Duplicate OrderIDs

✓ 0 Date Format Errors

RAW DATA AUDIT REPORT

Dataset Overview

Attribute	Value
Source File	Dataset_for_Data_Analytics_-_Sheet1.csv
Total Rows	1,200
Total Columns	14
Date Range	2023-01-04 to 2025-06-17
Products	Monitor, Phone, Tablet, Chair, Printer, Laptop, Desk (7 types)
Order Statuses	Shipped, Cancelled, Returned, Delivered, Pending
Payment Methods	Debit Card, Online, Credit Card, Gift Card, Cash

Column-Level Issue Scan

Column	Data Type	Null Count	Null %	Issue Detected
OrderID	str	0	0.0%	Clean
Date	str	0	0.0%	Type: String→Datetime
CustomerID	str	0	0.0%	Clean
Product	str	0	0.0%	Clean
Quantity	int64	0	0.0%	Clean
UnitPrice	float64	0	0.0%	Clean
ShippingAddress	str	0	0.0%	Clean
PaymentMethod	str	0	0.0%	Clean
OrderStatus	str	0	0.0%	Clean
TrackingNumber	str	0	0.0%	Clean
ItemsInCart	int64	0	0.0%	Clean
CouponCode	str	309	25.8%	Missing Values
ReferralSource	str	0	0.0%	Clean
TotalPrice	float64	0	0.0%	Clean

CHANGE LOG — FULL RECORD

PHASE 1: STRATEGIC IMPUTATION		Handle the Gaps. Don't just delete.
CR001	Missing Values	● Resolved
Column(s):	CouponCode	
Description:	Imputed 309 missing CouponCode values with "NONE" (no coupon applied)	
Method:	Constant Fill (Domain Knowledge)	
Impact:	Preserved 309 records instead of deleting rows	
PHASE 2: INTEGRITY AUDIT		One Truth, One Record.
CR002	Duplicate Check	● Verified Clean
Column(s):	All / OrderID	
Description:	Scanned for full-row duplicates (0 found) and duplicate OrderIDs (0 found)	
Method:	drop_duplicates() + groupby OrderID	
Impact:	Zero duplicates confirmed — data integrity verified	
PHASE 3: FORMAT STANDARDISATION		Speak One Language.
CR003	Data Type	● Resolved
Column(s):	Date	
Description:	Converted Date column from object/string to datetime64 (ISO 8601 YYYY-MM-DD)	
Method:	pd.to_datetime()	
Impact:	Enables date arithmetic, filtering, and time-series analysis	

CR004**Numeric Precision**
Resolved**Column(s):** UnitPrice, TotalPrice**Description:** Rounded UnitPrice and TotalPrice to 2 decimal places for currency consistency**Method:** `.round(2)`**Impact:** Standardised monetary values across all 1,200 records**CR005****Text Formatting**
Resolved**Column(s):** Product, OrderStatus, PaymentMethod, ReferralSource**Description:** Applied `.strip()` and `.title()` to remove whitespace and enforce Proper Case on categorical text columns**Method:** `str.strip() + str.title()`**Impact:** Prevents `groupBy` errors from case/space variants across 1,200 records**CR006****Text Formatting**
Resolved**Column(s):** CouponCode**Description:** Applied `.strip()` and `.upper()` to CouponCode for consistent uppercase coupon identifiers**Method:** `str.strip() + str.upper()`**Impact:** Ensures coupon grouping accuracy across all 1,200 records

BEFORE vs AFTER COMPARISON

Metric	RAW Dataset	CLEAN Dataset	Delta
Total Records	1,200	1,200	±0 (no rows deleted)
Missing Values	309	0	-309 imputed
Duplicate Rows	0	0	Verified clean
Duplicate OrderIDs	0	0	Verified clean
Date Format Errors	0	0	Verified clean
CouponCode Nulls	309	0	Filled → NONE
Date Column Type	object (string)	datetime64	Converted
Price Decimal Places	Variable	2 (rounded)	Standardised
Text Case Consistency	Mixed	Proper/Upper	Normalised

FINAL VERIFICATION — PROJECT 2 THRESHOLD GATE

Per DecodeLabs standard, the following assertions were run programmatically on the cleaned dataset before output. All checks passed with zero failures.

	Assertion	Result	Detail
✓	<code>assert df["OrderID"].duplicated().sum() == 0</code>	PASSED	0 duplicate IDs
✓	<code>assert df["Date"].isnull().sum() == 0</code>	PASSED	0 null dates
✓	<code>assert df.isnull().sum().sum() == 0</code>	PASSED	0 nulls anywhere
✓	<code>date format regex: YYYY-MM-DD on all 1,200</code>	PASSED	0% error rate

PROJECT 1 STATUS: COMPLETE

All cleaning phases executed. All verification gates passed.
Cleaned dataset ready for handoff to Project 2.